Lab Experiment 4

Global engineering collaboration problem (part 1):

Learning Outcomes

By the end of these activities, students will be able to:

1. Identify the differences between SI, BG, and USCS units commonly used in engineering practice.
2. Interpret mechanical drawings with missing or mixed unit systems.
3. Apply accurate unit conversions to complete engineering calculations and drawings.
4. Demonstrate problem-solving skills in real-world scenarios involving international collaboration.
5. Reflect on the importance of standardization and clear communication in global engineering projects.

Lab Exercise

Objective:

To simulate a real-world engineering scenario where students collaborate with an overseas company that uses a different system of units. Perform accurate conversions between SI and BG/USCS systems and complete the documentation to ensure clarity and accuracy.

Instructions:

Complete the following activities referring to chapter 4 and other appropriate resources in your lab report paper.

Lab activity overview:

You will be given an isometric drawing of each component in the delivery packet, as well as their technical specifications, **ignore the dimensions on the drawings when converting units.**

You work as an Executive Director in the engineering department of a logistics company, and were tasked to help one of the clients from abroad, here is the information your supervisor left:

E-MIDER, is a German-based manufacturer of vehicle subassemblies, they had just received permission to export mechanical devices to the U.S.

Your team is tasked with reviewing some component packages for North American customers who work primarily in Imperial Units. Once you familiarize yourself with the components, complete the following deliverables:

- **Create** a structured problem-solving approach for your team in the form of a list (Reference chapter 5 and the slides), to successfully complete the assigned tasks.
- **Read** and interpret the mechanical drawings, try to guess what component it is without looking at the technical specifications sheet (There are no right or wrong answers).
- **Identify** the line conventions used on **part2 drawing**, each one is numbered. Reference the number next to the answer in your lab report.
  For example:
  1. Visible Lines
  2. Section Lines
  3. Dimension Lines
- On the technical specifications document, **convert** the component specifications from **SI** to **BG** (Only convert specifications highlighted). If you used any online tool for conversion, please reference it in your lab report.

- Once you are done with the tasks above, modify the technical specifications file by improving the description of the component (**100 words min**.) and changing the original units to the converted ones. Submit the revised technical specifications sheet to Brightspace along with your lab report.

Reflection and transparency

Prepare a brief reflection that includes:

- The AI tool used and how (If you used one)
- What you learned while doing this lab

Grading Rubric

- **Objectives:** Identifies objectives and provides the necessary background or context (0-3 points)
- **Results/Analysis:** Presents results clearly. Analyzes findings correctly, with proper calculations and/or comparisons as needed. (0-6 points)
- **Discussion & Reflection:** Explains significance of results, discusses errors and connects outcomes to engineering/technology concepts. (0-5 points)
- **Professionalism & Format:** Report is well-organized, follows the provided template, uses clear language, proper grammar and formatting. Includes citation and references (if outside sources were used) (0-6 points)
- **Total:** 20 points

Deliverables

You must submit a lab report following the lab template. The lab report must include the **deliverables** on the activity problem, a **reflection** section as well as a **reference** section.
In addition, your submission must include the revised technical specifications sheet in **.pdf** format.